TCS NQT PYQ QUESTIONS

Company: TCS NQT
Round: Online
Year: 2025
Difficulty: Medium-Hard
Topic: Problems on Ages

Problem Description:

Five years ago, the ratio of A's age to B's age was 3:5. Ten years from now, the ratio of their ages will be 3:4.

Find A's present age.

Options:

A) 15 years
B) 18 years
C) 20 years
D) 24 years

Correct Answer:

C) 20 years

Approach to Solve This Problem:

STEP 1:

Let A's age 5 years ago = 3k and B's age 5 years ago = 5k, for some positive number k.

So the present ages are: A = 3k + 5, and B = 5k + 5.

STEP 2:

Ten years from now: A's age = 3k + 15, and B's age = 5k + 15.

Given that this ratio is 3:4, we write: (3k + 15) / (5k + 15) = 3/4.

STEP 3:

Cross-multiplying: 4(3k + 15) = 3(5k + 15), which gives 12k + 60 = 15k + 45.

So 15 = 3k, hence k = 5.

STEP 4:

A's present age = 3k + 5 = 3(5) + 5 = 20 years.

CORRECT ANS = 20 years